Project Handbook: Simulating Decoherence of Schrödinger Cat States via Quantum Monte Carlo Wavefunctions

A 5-Week Implementation, Benchmarking, and Physics Guide

Numerical Physics Course — Comprehensive Project Specification

Academic Year 2026/2027

Executive Summary & Project Mission

This document provides the complete theoretical, numerical, and algorithmic roadmap for a 5-week computational physics project. The core objective is to model the quantum-to-classical transition of macroscopic superpositions (**Schrödinger cat states**) subject to environmental dissipation. Rather than relying solely on the $\mathcal{O}(N^2)$ deterministic Lindblad master equation, the dynamics are simulated using the $\mathcal{O}(N)$ stochastic **Monte Carlo Wavefunction (MCWF)** method (quantum-jump approach). Key deliverables include: verification of $1/\sqrt{M}$ Monte Carlo convergence against exact Lindblad integration, phase-space Wigner distribution animations, validation of the accelerated decoherence scaling law $\tau_{\text{dec}} \approx \tau_{\text{diss}}/(2|\alpha|^2)$ (Brune et al., 1996), and single-trajectory quantum jump parity flips.

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1 Theoretical Foundations & The Physical System

1.1 The Single-Mode Bosonic Cavity in Fock Space

We consider a single bosonic mode (e.g., an optical or superconducting microwave cavity) described in the photon Fock basis $\{|n\rangle\}_{n=0}^{N_{\text{cut}}-1}$. The Hilbert space is truncated at a finite cutoff N_{cut}, chosen sufficiently large such that the probability of occupying states near N_{cut} is negligible ($\langle N_{\text{cut}}-1|\psi\rangle < 10^{-7}$).

The fundamental ladder operators satisfy $[a, a^\dagger] = \mathbb{I}$, defined by:

$$a|n\rangle = \sqrt{n}\,|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \quad \hat{n} = a^\dagger a, \quad \hat{n}|n\rangle = n|n\rangle. \tag{1}$$

In an N_{cut}-dimensional matrix representation:

$$a_{m,n} = \sqrt{n}\,\delta_{m,n-1}, \quad (a^\dagger)_{m,n} = \sqrt{m}\,\delta_{m,n+1}. \tag{2}$$

1.2 Macroscopic Quantum Superpositions: Schrödinger Cat States

A coherent state $|\alpha\rangle$ ($\alpha \in \mathbb{C}$) is an eigenstate of the annihilation operator $a|\alpha\rangle = \alpha|\alpha\rangle$, expanded in Fock space as:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \tag{3}$$

A **Schrödinger cat state** is a quantum superposition of two distinct coherent states with opposite phases $|\alpha\rangle$ and $|-\alpha\rangle$:

$$|\psi_+\rangle = \mathcal{N}_+ \left(|\alpha\rangle + |-\alpha\rangle\right) = 2\mathcal{N}_+ e^{-|\alpha|^2/2} \sum_{k=0}^{\infty} \frac{\alpha^{2k}}{\sqrt{(2k)!}} |2k\rangle \quad \text{(Even Cat)}, \tag{4}$$

$$|\psi_-\rangle = \mathcal{N}_- \left(|\alpha\rangle - |-\alpha\rangle\right) = 2\mathcal{N}_- e^{-|\alpha|^2/2} \sum_{k=0}^{\infty} \frac{\alpha^{2k+1}}{\sqrt{(2k+1)!}} |2k+1\rangle \quad \text{(Odd Cat)}, \tag{5}$$

with normalization factors given by:

$$\mathcal{N}_\pm = \frac{1}{\sqrt{2\left(1 \pm e^{-2|\alpha|^2}\right)}}. \tag{6}$$

Crucial Property: An even cat state contains *only even photon numbers* ($n = 0, 2, 4, \dots$), whereas an odd cat state contains *only odd photon numbers* ($n = 1, 3, 5, \dots$).

1.3 Dissipative Channels: Photon Loss vs. Pure Dephasing

The interaction with an external thermal reservoir in the Markovian limit is governed by Lindblad jump operators:

1. **Amplitude Damping (Photon Loss):** $L = \sqrt{\kappa}\,a$, where $\kappa = 1/\tau_{\text{diss}}$ is the photon decay rate.
2. **Pure Dephasing (Phase Jitter):** $L_\phi = \sqrt{\gamma}\,a^\dagger a$, which scrambles the relative phase between Fock states without altering total cavity energy.

Key Physical Insight: Parity Flips under Photon Loss

Notice what happens when a single photon is lost from an even cat state:

$$a|\psi_+\rangle \propto a \sum_k c_{2k}|2k\rangle = \sum_k c_{2k}\sqrt{2k}|2k-1\rangle \propto |\psi_-\rangle. \tag{7}$$

Each discrete photon jump flips an even cat state into an odd cat state! This fundamental quantum jump effect is directly observable in individual MCWF trajectories.

2 The Two Numerical Frameworks

2.1 Method 1: Direct Master Equation (Lindblad)

The unconditional time evolution of the system's density matrix $\rho(t)$ is given by the Lindblad master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_0, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right), \tag{8}$$

where $\{A, B\} = AB + BA$. In the interaction picture ($H_0 = 0$) with photon loss:

$$\frac{d\rho}{dt} = \kappa \left(a\rho a^\dagger - \frac{1}{2} a^\dagger a \rho - \frac{1}{2} \rho a^\dagger a \right). \tag{9}$$

Computational Cost: Storing $\rho(t)$ requires N_{cut}^2 elements. Vectorizing ρ into an N_{cut}^2-dimensional vector $\vec{\rho}$ yields a Liouvillian superoperator $\mathcal{L}$ of size $N_{\text{cut}}^2 \times N_{\text{cut}}^2$. For $N_{\text{cut}} = 60$, $\mathcal{L}$ contains $(3600)^2 \approx 1.3 \times 10^7$ matrix elements.

2.2 Method 2: Monte Carlo Wavefunction (MCWF / Quantum Jumps)

Introduced by Dalibard, Castin, and Mølmer (1992) and Carmichael (1993), the MCWF method unravels the master equation into stochastic trajectories of pure state vectors $|\psi(t)\rangle$ of size N_{cut} ($\mathcal{O}(N)$ memory scaling).

2.2.1 The Algorithmic Workflow

1. **Effective Hamiltonian:** Between jumps, the non-unitary evolution is generated by the non-Hermitian operator:

$$H_{\text{eff}} = H_0 - \frac{i\hbar}{2} \sum_k L_k^\dagger L_k = -\frac{i\hbar\kappa}{2} a^\dagger a. \tag{10}$$

   The Schrödinger equation under H_{eff} reads:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H_{\text{eff}} |\psi(t)\rangle. \tag{11}$$

   Because H_{eff} is non-Hermitian, the norm squared $P(t) = \langle\psi(t)|\psi(t)\rangle$ strictly decreases monotonically over time.

2. **Stochastic Jump Criterion:**

   - At the start of a trajectory (or immediately following a jump at time t_0), sample a random number $r \sim \mathcal{U}(0, 1)$ from a uniform distribution.
   - Integrate the state forward in time using a 4th-order Runge-Kutta (RK4) scheme.
   - At each step t, compute the norm $P(t) = \langle\psi(t)|\psi(t)\rangle$.
   - A quantum jump occurs at the precise time t^* where:

$$P(t^*) = r. \tag{12}$$

3. **Jump Execution and State Reset:** When $P(t) \leq r$:

   - Apply the jump collapse operator: $|\psi(t^*)\rangle \longleftarrow \frac{L|\psi(t^*)\rangle}{\|L|\psi(t^*)\rangle\|} = \frac{a|\psi(t^*)\rangle}{\|a|\psi(t^*)\rangle\|}$.
   - Re-normalize the state vector: $\langle\psi(t^*)|\psi(t^*)\rangle = 1$.
   - Draw a new random threshold $r' \sim \mathcal{U}(0, 1)$ and resume RK4 integration.

4. **Ensemble Averaging:** Repeat for M independent stochastic trajectories $\{|\psi^{(m)}(t)\rangle\}_{m=1}^M$. The physical density matrix is reconstructed via:

$$\bar{\rho}_M(t) = \frac{1}{M} \sum_{m=1}^{M} |\psi^{(m)}(t)\rangle\langle\psi^{(m)}(t)|. \tag{13}$$

3 Observables & Physical Metrics to Compute

3.1 1. Reconstructed Density Matrix $\bar{\rho}(t)$

Reconstructed on the Fock grid according to Eq. (13). The diagonal elements $\rho_{nn}(t)$ provide the photon-number probability distribution.

3.2 2. Purity $\mathcal{P}(t)$

$$\mathcal{P}(t) = \mathrm{Tr}\left(\bar{\rho}(t)^2\right). \tag{14}$$

Expected Behavior:

- At $t = 0$: $\mathcal{P}(0) = 1.0$ (initial pure cat state).
- At intermediate times $\tau_{\mathrm{dec}} \ll t \ll \tau_{\mathrm{diss}}$: $\mathcal{P}(t) \to 0.5$ (incoherent statistical mixture $\frac{1}{2}|\alpha\rangle\langle\alpha| + \frac{1}{2}|-\alpha\rangle\langle-\alpha|$).
- At long times $t \gg \tau_{\mathrm{diss}}$: $\mathcal{P}(t) \to 1.0$ (relaxation to the pure vacuum state $|0\rangle$).

3.3 3. Parity Observable $\langle\hat{\Pi}(t)\rangle$

The parity operator is defined as $\hat{\Pi} = (-1)^{a^\dagger a} = \exp(i\pi a^\dagger a)$:

$$\langle\hat{\Pi}(t)\rangle = \mathrm{Tr}\left(\bar{\rho}(t)\hat{\Pi}\right) = \sum_{n=0}^{N_{\mathrm{cut}}-1} (-1)^n \rho_{nn}(t). \tag{15}$$

For an even cat state, $\langle\hat{\Pi}(0)\rangle = +1$. For an odd cat state, $\langle\hat{\Pi}(0)\rangle = -1$.

3.4 4. The Phase-Space Wigner Function $W(x, p)$

The Wigner quasi-probability distribution represents the quantum state in classical phase space (x, p). For a density operator ρ:

$$W(x,p) = \frac{1}{\pi\hbar}\int_{-\infty}^{\infty} \langle x - y|\rho|x + y\rangle e^{2ipy/\hbar} dy. \tag{16}$$

In dimensionless quadrature coordinates ($x = \mathrm{Re}(\beta)\sqrt{2}$, $p = \mathrm{Im}(\beta)\sqrt{2}$ with $\beta \in \mathbb{C}$), $W(\beta)$ can be computed numerically via displaced parity:

$$W(\beta) = \frac{2}{\pi}\mathrm{Tr}\left[D^\dagger(\beta)\,\rho\,D(\beta)\,\hat{\Pi}\right], \tag{17}$$

where $D(\beta) = \exp(\beta a^\dagger - \beta^* a)$ is the Glauber displacement operator.

Fundamental Mathematical Identity

Evaluating the Wigner distribution at the origin of phase space ($\beta = 0$) gives:

$$W(0,0) = \frac{2}{\pi}\mathrm{Tr}[\rho\hat{\Pi}] = \frac{2}{\pi}\langle\hat{\Pi}\rangle. \tag{18}$$

The central value of the Wigner function is directly proportional to the expectation value of photon parity!

3.5 5. Exact Analytical Benchmarks & Decoherence Timescale

According to the landmark theory verified by Brune et al. (1996), the off-diagonal coherence terms of a cat state decaying into a zero-temperature bath behave analytically as:

$$\rho_{+-}(t) \propto \exp\left[-2|\alpha|^2\left(1-e^{-\kappa t}\right)\right] \approx \exp\left(-2|\alpha|^2\kappa t\right) \quad (\text{for } \kappa t \ll 1). \tag{19}$$

Therefore, the characteristic decoherence time τ_{dec} satisfies:

$$\tau_{\text{dec}} = \frac{1}{2\kappa|\alpha|^2} = \frac{\tau_{\text{diss}}}{2|\alpha|^2}. \tag{20}$$

For $|\alpha| = 2.5$, $\tau_{\text{dec}} = \tau_{\text{diss}}/12.5$: quantum interference vanishes **over an order of magnitude faster** than photon dissipation!

4 Five-Week Implementation Blueprint

Week	Objective	Key Tasks & Algorithms	Verification Benchmark
Week 1	**Fock Basis & State Generation**	• Construct matrices $a, a^\dagger, \hat{n}, \hat{\Pi}$ for $N_{\text{cut}} \approx 40$. • Generate $\|\alpha\rangle$ and even/odd cats. • Implement Wigner calculation (Eq. 17).	• Coherent state: $W \geq 0$ everywhere. • Cat state: $W(0,0) = \pm 2/\pi$ with deep negative interference ripples. • Orthogonality: $\langle\psi_+|\psi_-\rangle = 0$.
Week 2	**The MCWF Engine**	• Implement RK4 solver for $H_{\text{eff}} = -\frac{i\hbar\kappa}{2}a^\dagger a$. • Build jump detector $P(t) \leq r$. • Implement state collapse and normalization.	• Single trajectory inspection: discrete norm collapses. • Verify that each jump flips photon parity ($\langle\Pi\rangle \to -\langle\Pi\rangle$).
Week 3	**Lindblad Solver & Ensemble Benchmark**	• Implement direct Lindblad ODE solver (Eq. 8). • Run M parallel MCWF trajectories. • Reconstruct $\bar{\rho}_M(t)$.	• Measure Frobenius error $\|\bar{\rho}_M(t) - \rho_{\text{Lindblad}}(t)\|_F$. • Benchmark plot: prove error scales as $\mathcal{O}(M^{-1/2})$.
Week 4	**Decoherence Scaling Analysis**	• Compute $\mathcal{P}(t)$, $S(t)$, and $\langle\Pi(t)\rangle$. • Sweep cat amplitudes: $\alpha \in [1.0, 3.0]$. • Fit decay rate of $W(0,0,t)$ to extract $\tau_{\text{dec}}(\alpha)$. • Compare photon loss vs dephasing.	• Log-log plot of τ_{dec} vs $\|\alpha\|$ yielding a slope of -2.0 ± 0.05 (validating Eq. 20 and Brune et al. 1996).
Week 5	**Visual Suite & Final Synthesis**	• Generate phase-space animations of $W(x,p,t)$. • Render side-by-side: single jump trajectory vs. smooth ensemble. • Assemble final report and code repository.	• Visual verification of the two Gaussian peaks remaining while central interference ripples vanish at $t \sim \tau_{\text{dec}}$.

Table 1: Detailed 5-Week Milestone and Deliverables Schedule.

5 Core Algorithmic Recipes (Pseudocode)

5.1 Algorithm 1: Single MCWF Trajectory Integrator

```
import numpy as np

def run_mcwf_trajectory(psi0, H_eff, L_jump, t_eval, dt):
    """
␣␣␣␣Simulates␣a␣single␣stochastic␣quantum␣jump␣trajectory.
␣␣␣␣"""
    state = psi0.copy()
    states_out = []
    r = np.random.uniform(0.0, 1.0)

    for t in t_eval:
        states_out.append(state / np.linalg.norm(state))

        # RK4 step for non-Hermitian H_eff
        k1 = -1j * H_eff @ state
        k2 = -1j * H_eff @ (state + 0.5 * dt * k1)
        k3 = -1j * H_eff @ (state + 0.5 * dt * k2)
        k4 = -1j * H_eff @ (state + dt * k3)
        state_next = state + (dt / 6.0) * (k1 + 2*k2 + 2*k3 + k4)

        # Norm test
        norm_sq = np.vdot(state_next, state_next).real
        if norm_sq <= r:
            # Quantum Jump Event!
            collapsed = L_jump @ state_next
            state = collapsed / np.linalg.norm(collapsed)
            r = np.random.uniform(0.0, 1.0) # Reset threshold
        else:
            state = state_next

    return np.array(states_out)
```

5.2 Algorithm 2: Wigner Function Calculation via Displaced Parity

```
import scipy.linalg as la

def compute_wigner_grid(rho, a_op, xvec, pvec):
    """
␣␣␣␣Computes␣W(x,␣p)␣over␣a␣2D␣grid␣using␣the␣displaced␣parity␣operator␣identity.
␣␣␣␣"""
    N = a_op.shape[0]
    parity = np.diag([(-1)**n for n in range(N)])
    W = np.zeros((len(xvec), len(pvec)))

    for i, x in enumerate(xvec):
        for j, p in enumerate(pvec):
            beta = (x + 1j * p) / np.sqrt(2.0)
            # Displacement operator: D(beta) = exp(beta * a^dagger - beta^* * a)
            D = la.expm(beta * a_op.conj().T - np.conj(beta) * a_op)
            rho_displaced = D.conj().T @ rho @ D
            W[i, j] = (2.0 / np.pi) * np.trace(rho_displaced @ parity).real

    return W
```

6 Essential Literature References & Their Key Results

References

[1] **J. Dalibard, Y. Castin, and K. Mølmer**, *Wave-function approach to dissipative processes in quantum optics*, **Physical Review Letters 68**, 580 (1992).

Foundational Result: Introduces the Monte Carlo Wavefunction method. Demonstrates that averaging M pure state trajectories evolved under $H_{\text{eff}} = H - \frac{i\hbar}{2}\sum L_k^\dagger L_k$ with stochastic collapses reproduces the exact Lindblad master equation with $\mathcal{O}(N)$ memory scaling rather than $\mathcal{O}(N^2)$.

[2] **K. Mølmer, Y. Castin, and J. Dalibard**, *Monte Carlo wave-function method in quantum optics*, **Journal of the Optical Society of America B 10**, 524–538 (1993).

Foundational Result: The comprehensive tutorial and practical guide. Section II gives the explicit step-by-step algorithmic recipe for jump detection and proves the mathematical equivalence to order Δt.

[3] **M. Brune, E. Hagley, J. Dreyer, X. Maître, A. Maali, C. Wunderlich, J. M. Raimond, and S. Haroche**, *Observing the Progressive Decoherence of the "Meter" in a Quantum Measurement*, **Physical Review Letters 77**, 4887 (1996).

Landmark Experimental Result (2012 Nobel Prize): First real-time observation of the decoherence of a Schrödinger cat state in a superconducting microwave cavity. Formulates the exact decoherence timescale $\tau_d = \frac{T_{\text{cav}}}{2D^2} = \frac{1}{2\kappa|\alpha|^2}$ and the two-atom fringe contrast decay $\eta(\tau) = \frac{1}{2}\cos\phi[1 + \exp(-\tau/\tau_d)]$.

[4] **M. B. Plenio and P. L. Knight**, *The quantum-jump approach to dissipative dynamics in quantum optics*, **Reviews of Modern Physics 70**, 101–144 (1998).

Definitive Review: Over 2,000 citations. Detailed analysis of waiting-time distributions $P_0(\tau) = \langle\psi_0|e^{iH_{\text{eff}}^\dagger\tau/\hbar}e^{-iH_{\text{eff}}\tau/\hbar}|\psi_0\rangle$, non-Hermitian dynamics, and dark-state dynamics.

[5] **W. H. Zurek**, *Decoherence, einselection, and the quantum origins of the classical*, **Reviews of Modern Physics 75**, 715–775 (2003).

Theoretical Foundation: Explains the concept of environment-induced superselection (einselection), showing why coherent states form the stable classical "pointer states" and why macroscopic superpositions rapidly degrade into classical statistical mixtures.